Anna French

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U.S. Permanent Resident

PLACEMENT CONTACTS

Kelly Bishop, Placement Director
kelly.bishop@asu.edu

Laura Talts, Placement Coordinator
ltalts@asu.edu

EDUCATION

Arizona State University *Fall 2021–expected May 2027*
PhD in Economics

New York University *Spring 2025*
Visiting PhD Student

Arizona State University *Fall 2021–Fall 2022*
M.S. in Economics

New Economic School and Higher School of Economics *Fall 2017–Spring 2021*
B.A. in Economics (double degree)

RESEARCH INTERESTS

Macroeconomics; Environmental and Energy Economics; Spatial Economics

REFERENCES

Wyatt Brooks (Advisor)
Associate Professor
Arizona State University
wyattjbrooks@gmail.com

Nicolai Kuminoff
Associate Professor
Arizona State University
kuminoff@asu.edu

Nicholas Vreugdenhil
Assistant Professor
Arizona State University
nvreugde@asu.edu

WORKING PAPERS

"The Aggregate Impact of Policy Uncertainty in the Green Energy Transition" **JOB MARKET PAPER**

Abstract: Federal wind and solar subsidies are central to U.S. electricity-sector decarbonization, yet they have been historically unstable. I develop a dynamic spatial general-equilibrium model of the U.S. economy and electricity sector with stochastic subsidies and irreversible electricity capital, and use it to quantify the cost of this instability. I compare counterfactual transition paths under a stable subsidy, available in every year, and under a budget-neutral unstable subsidy whose availability changes at dates firms do not know in advance. Under the stable subsidy, renewable capacity is 13% higher by 2070 and gas capacity is lower, avoiding $360 billion in climate damages and $46 billion in mortality damages from local air pollution. I find that the stable subsidy generates $292 billion in net benefits. The effects are largest in regions such as the Midwest, where the subsidy determines whether new renewable capacity is built. Perfect-foresight counterfactuals show that these gains come from eliminating policy switching itself; resolving uncertainty about the timing of policy changes while the subsidy still switches would lower welfare.

"Spatial Effects of Diesel Tax: Trade and Externalities" — Submitted to the *Quarterly Journal of Economics*

Abstract: Freight transported by heavy-duty trucks generates carbon, local pollution, congestion, accident, and pavement damages along their routes. I develop a multi-commodity open-economy trade model that follows each trade flow through the U.S. highway network, allocating emissions, damages, and tax revenue to every state and county it crosses. Using Freight Analysis Framework shipment data and web-scraped diesel prices, I estimate freight responses to fuel costs. Increasing the federal diesel tax to its 1993 real value avoids more than $700 million in damages over 2018–2024, with pavement damage generating the largest benefit. Most benefits occur in corridor states that freight passes through, and part of the welfare cost is borne by foreign buyers of U.S. exports. A unilateral state diesel tax creates large spillovers: when New Mexico raises its tax to California's level, freight shifts to alternative corridors, increasing externalities in neighboring states.

"Permits as Real Options: Evidence and Implications for Regulatory Leakage"
(with Nicholas Vreugdenhil) — Submitted to the *American Economic Review*

Abstract: We identify a new channel for regulatory leakage: the "permitting channel." In many settings, permits lock in the regulatory regime at approval, allowing firms to acquire the option to invest under current rules. Unlike the conventional Green Paradox response, this channel operates without accelerating physical investment and pollution. We develop a framework in which firms can use real options to avoid anticipated regulation. We estimate the framework using Colorado's oil and gas reforms, where permits surged before regulation but drilling did not. Policy-relevant designs that remove the permitting channel substantially reduce the health risks from pollution despite amplifying anticipatory investment.

WORK IN PROGRESS

"Environmental Policy, Local Externalities and Development: Evidence from the Amazon" (with Wyatt Brooks)

Abstract: The Amazon Rainforest provides substantial environmental benefits but also occupies land with valuable productive uses. We study the local economic effects of restricting deforestation when land-intensive agriculture imposes negative externalities on other economic activity. We develop a model in which households allocate labor between agriculture and fishing, and deforestation for agriculture reduces fishing productivity through soil erosion and runoff. Empirically, we exploit the asymmetric relaxation of environmental enforcement during the Bolsonaro administration, which affected federal but not state jurisdictions. Using a spatial regression discontinuity design across federal-state boundaries, we find that deforestation increased significantly more within federal reserves. Novel village-level data show that this deregulation reduced local income. The gains from agricultural expansion are outweighed by productivity losses in fishing, implying that deforestation restrictions can generate both environmental and local economic benefits.

AWARDS AND FELLOWSHIPS

AEA Summer Fellowship, Washington Center for Equitable Growth	*Summer 2026*
Best Progress Towards Dissertation Award, Arizona State University	*Fall 2025*
Most Deserving Graduate Student Award, Arizona State University	*Spring 2024*
Best Oral Presentation Award (Comprehensive Exam), Arizona State University	*Spring 2024*
Travel Fellowship, Environmental and Energy Economics Summer School, UC Berkeley	*Summer 2023*
Best Macroeconomics Qualifying Exam Award, Arizona State University	*Summer 2022*
Outstanding Female First-Year Doctoral Fellowship, Arizona State University	*Fall 2021*
National Winner (top 12 of 10,000+), All-Russian Olympiad in Economics	*Spring 2017*

PRESENTATIONS

2026 Midwest Macroeconomics Meeting (Texas Tech University)*; ASU Economics PhD Reunion Conference (Arizona State University)*; Arizona Economics Mini-Conference (University of Arizona); Modeling Climate-Dynamics Workshop for Young Scholars (University of Chicago)

2025 Midwest Macroeconomics Meeting (Federal Reserve Bank of Kansas City); Environmental & Resource Economics Workshop (University of Colorado); AZ Workshop on Environment, Natural Resource, and Energy Economics (University of Arizona); Center for Environmental Economics and Sustainability Policy Seminar (Arizona State University)

2024 Midwest Macroeconomics Meeting (Virginia Commonwealth University)

2023 AZ Workshop on Environment, Natural Resource, and Energy Economics (Arizona State University)

Internal: ASU Macroeconomics Workshop (2024–2026); ASU General Economics Workshop (2023)

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TEACHING EXPERIENCE

Instructor, Arizona State University *Fall 2025*
Intermediate Microeconomic Theory (ECN 312)

Teaching Assistant, Arizona State University *Fall 2022–Spring 2026*
Environmental Economics (ECN 445; Fall 2022, Spring 2023), Microeconomic Principles (ECN 212; Fall 2023, Spring 2024), Economics of Gender & Family (ECN 349; Spring 2026)

PRE-DOCTORAL TEACHING

Instructor and Homeroom Teacher, School No. 57, Moscow *2017–2021*
One of Russia's leading high schools in mathematics. Taught a university-level economics curriculum; 6 students (one third of the class) won national awards at the All-Russian High School Olympiad in Economics. Helped build the mathematics–economics track from scratch, led 8th-grade admissions for four years, and designed and directed a two-week camp for 100 students, hiring and managing its instructors.

Olympiad Economics Coach, Moscow Team and National Training Camps *2017–2021*
Coached Moscow's team, the strongest in the country, for the national final of the All-Russian High School Olympiad in Economics, and taught at 10+ intensive training camps across Russia.

RESEARCH EXPERIENCE

Research Assistant, Arizona State University *Summer 2023–present*
Wyatt Brooks (Summer 2023–present); Nicholas Vreugdenhil (Summers 2024–2025); Nicolai Kuminoff (Summer 2024); Kelly Bishop (Summer 2023)

Research Assistant, Tufts University *Spring 2019*
Christopher R. Miller

Research Assistant, New Economic School *Fall 2018–Fall 2019*
Andrei Markevich

OTHER INFORMATION

Programming Skills: Julia, Python, R, Stata, MATLAB, LaTeX
Languages: English (fluent), Russian (native), French (intermediate)
Former Name: Anna Samodelkina